Power and a staged analysis plan for the Proposition 40 survey experiment

Fernando Hoces de la Guardia
Pre-analysis plan, appendix on power

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1 Why this appendix

We do not know how many people will answer the survey. It may be a few dozen or a few hundred. The plan therefore registers one main hypothesis with one outcome, which is tested whatever the number of responses. It also registers further analyses, each run only if the realized number of responses is large enough for that analysis to detect an effect of a stated size. This appendix gives the formulas behind those thresholds, checks them with simulations of the registered estimator, and states the rule that decides which analyses are run.

Every number below is computed by `comparison/survey/power/sim.py` and read into this document from its output. None is typed by hand.

2 Summary: responses needed to detect 0.3 standard deviations

Table 1 gives the smallest number of responses at which each analysis has 80% power against an effect of 0.3 standard deviations, at the 5% level with the Romano-Wolf adjustment inside each family (Section 4). Subgroups have equal shares. Column N^* is the Romano-Wolf threshold from the simulations of Section 9.1. Column Bound is the closed-form Bonferroni bound, which is at or above it. For the five questions, column A is an effect of 0.3 standard deviations on one question. Column B is the smaller effect on each question, about 0.18, that goes with an effect of 0.3 on the index (equation (7)). For heterogeneity, column A is the effect inside one cell and column B is the highest-order difference of effects (Section 6). Each family covers every combination of the $D = 3$ candidate dimensions at that order.

		A			B		
	Analysis	Tests m	N^*	Bound	Tests m	N^*	Bound
1	Main hypothesis H1: the lean index	1	349				
2	H1 by outcome, each of the five questions. A: 0.3 SD on a question. B: the question-level effect that matches 0.3 SD on the index	5	410	520	5	1,130	1,433
3	Heterogeneity, one dimension at a time (3 combinations). A: the effect inside each of 2 cells. B: the difference between the two halves	6	890	1,077	3	1,862†	1,862
4	Heterogeneity, two dimensions at a time (3 combinations). A: the effect inside each of 4 cells. B: the 2-way difference	12	1,960	2,443	3	7,445†	7,445
5	Heterogeneity, three dimensions at a time (1 combination). A: the effect inside each of 8 cells. B: the 3-way difference	8	3,670	4,547	1	22,326	

Table 1: Responses needed for 80% power against an effect of 0.3 standard deviations, equal shares. N^*: the simulated Romano-Wolf threshold, rounded up to the next 10. Bound: the Bonferroni bound from equations (3), (7) and (6). A single test ($m = 1$) needs no adjustment, so its N^* comes from the formula. †The simulation found no Romano-Wolf threshold below the bound, so the bound is kept.

3 Setup

Respondent $i = 1, \ldots, N$ is assigned to treatment, $T_i = 1$, or control, $T_i = 0$, with probability 1/2, within three entry-point strata. On each of the $K = 5$ required questions the answer is coded

$$y_{ik} = \begin{cases} +1 & \text{the BSZ value} \\ -1 & \text{the JR value} \\ 0 & \text{about equally credible, or unsure.} \end{cases}$$

The primary outcome is the lean index, $\text{Lean}_i = \frac{1}{K}\sum_{k=1}^{K} y_{ik}$. The main hypothesis is tested with

$$\text{Lean}_i = a + b\,T_i + \textstyle\sum_s g_s E_{s,i} + e_i, \qquad H_1 \colon b = 0 \text{ against } b \neq 0, \tag{1}$$

with entry-point indicators $E_{s,i}$ and HC2 standard errors.

Effects are stated in standard-deviation units of the outcome in the control group: a standardized effect $d = b/\sigma$. This avoids guessing the outcome's spread, which we cannot know before the data arrive.

4 Multiple testing

Families. The main hypothesis H1 is one test and is not adjusted. Every other analysis is a family of tests, and each family is adjusted on its own. The families are the five question-level tests, the questions inside each half, the effects inside the cells (one family per order), the one-dimension differences (H2 to H4), and the two-way differences (Section 6). The three-way difference is a single test.

The procedure. Each family is adjusted with the Romano-Wolf step-down procedure (Romano and Wolf, 2005), computed as in Clarke, Romano, and Wolf (2020). It works with studentized statistics, $t_j = \hat{b}_j / \widehat{\mathrm{se}}_j$, from the registered equations with HC2 standard errors. A pairs bootstrap redraws respondents with replacement within each entry-point stratum, keeping each stratum's size. Every regression of the family is re-estimated on the same bootstrap sample. Draw r gives $t^*_{j,r} = |\hat{b}^*_{j,r} - \hat{b}_j| / \widehat{\mathrm{se}}^*_{j,r}$. Centering at the estimate makes the draws mimic the statistics when every null hypothesis is true. Order the tests so that $|t_{(1)}| \geq \cdots \geq |t_{(S)}|$. With B draws, the adjusted p-value of test (s) is

$$p_{(s)} = \max_{s' \leq s} \frac{1 + \#\{r : \max_{j \geq s'} t^*_{(j),r} \geq |t_{(s')}|\}}{1 + B}.$$

The first test is compared with the maximum over all tests, the second with the maximum over all tests but the first, and so on down the list. A test is rejected when its adjusted p-value is at most 5%. The registered analysis uses $B = 2{,}000$ draws.

Why it fits. The procedure uses the joint distribution of the statistics, so it charges less for tests that move together. The five questions share one latent lean, with an average correlation of 0.24 in the simulated population of Section 9. The cells of different dimensions share respondents. Bonferroni's method, and Holm's method that an earlier draft of this appendix used, ignore this correlation and so give up power. When the tests are close to independent, as the differences across independent dimensions are, Romano-Wolf gains little.

Bounds. In large samples, the Romano-Wolf critical value at the first step is at most the Bonferroni critical value $z_{1-\alpha/(2m)}$, and later steps use smaller ones. A threshold computed with the Bonferroni critical value is therefore an upper bound on the Romano-Wolf threshold. The formulas below use this bound because it has a closed form. Section 9.1 simulates the Romano-Wolf thresholds themselves.

5 Analytic minimum detectable effects

Let $M(m) = z_{1-\alpha/(2m)} + z_{1-\kappa}$ for a two-sided test at level $\alpha = 0.05$ with power $1 - \kappa = 0.80$, where m is the number of tests in the family. The first term is the Bonferroni critical value, so $M(m)$ gives upper bounds on the Romano-Wolf thresholds (Section 4). Numerically $M(1) = 2.80$, $M(3) = 3.24$ and $M(5) = 3.42$.

Main effect. With half of N treated, $\mathrm{Var}(\hat{b}) = \sigma^2 \left(\frac{1}{N/2} + \frac{1}{N/2} \right) = 4\sigma^2/N$, so

$$\mathrm{MDE}_{\mathrm{main}}(N, m) = \frac{2\, M(m)}{\sqrt{N}}. \tag{2}$$

With $N = 50, 100, 200, 400$ responses the main test detects standardized effects of 0.79, 0.56, 0.40 and 0.28.

Threshold. Setting the MDE equal to a target d^* and solving for N gives the number of responses the analysis needs:

$$N^*_{\mathrm{main}}(d^*, m) = \left(\frac{2M(m)}{d^*} \right)^2. \tag{3}$$

6 Heterogeneity by order

The candidate dimensions. The plan declares $D = 3$ binary dimensions, all recorded before treatment: expertise (Yes or Somewhat, against No), prior exposure (No, against Yes or Not sure), and entry point (the explorer button, against the landing panel or the share link). Whether a respondent gives an email is asked after treatment. It is descriptive and is not one of the D. Any analysis of heterogeneity picks some of the D dimensions, and the more it picks, the more combinations there are to test and the thinner each cell becomes. The thresholds below count both.

Cells. An analysis of order k takes k of the D dimensions. There are $\binom{D}{k}$ such combinations. Crossing the k dimensions gives 2^k cells c, with shares s_c of the sample. The treatment effect inside cell c, τ_c, is estimated on $s_c N$ responses, half treated, so

$$\mathrm{Var}(\hat{\tau}_c) = \frac{4\sigma^2}{N s_c}. \tag{4}$$

The k-way difference. The highest-order interaction of a combination is the alternating sum of the cell effects,

$$\Delta_k = \sum_c (-1)^{\,k-|c|}\,\tau_c,$$

where $|c|$ is the number of the k dimensions at 1 in cell c. For $k = 1$ this is $\tau_1 - \tau_0$, the difference between two subgroups' effects. For $k = 2$ it is $(\tau_{11} - \tau_{10}) - (\tau_{01} - \tau_{00})$, a difference of differences. In the saturated regression with 0/1 indicators, Δ_k is the coefficient on $T_i \times G_{1i} \times \cdots \times G_{ki}$. The cells hold different respondents, so the estimates are independent and

$$\mathrm{Var}(\hat{\Delta}_k) = \sum_c \frac{4\sigma^2}{N s_c}. \tag{5}$$

With equal shares, $s_c = 2^{-k}$, the sum has 2^k terms of 2^k each:

$$\mathrm{Var}(\hat{\Delta}_k) = 4^k\,\frac{4\sigma^2}{N}, \qquad \mathrm{MDE}_{\Delta_k} = 2^k\,\frac{2M(m)}{\sqrt{N}}.$$

Each added dimension doubles the smallest detectable difference, or multiplies the responses needed by four. If the dimensions are independent with shares p_j, the sum factors:

$$\sum_c \frac{1}{s_c} = \prod_{j=1}^{k} \frac{1}{p_j(1-p_j)}.$$

For $k = 1$ this is the familiar $1/(p(1-p))$.

Families. The tests of one order form one Romano-Wolf family. For the cell effects of order k the family has $m = \binom{D}{k}2^k$ tests: every cell of every combination. For the k-way differences it has $m = \binom{D}{k}$ tests. With $D = 3$, the cell families have 6, 12 and 8 tests for $k = 1, 2, 3$, and the difference families have 3, 3 and 1.

Thresholds. Setting each MDE equal to d^* gives

$$N^*_{\text{cell}}(k) = \frac{1}{\min_c s_c}\left(\frac{2M(m)}{d^*}\right)^2, \qquad N^*_{\Delta}(k) = \left(\sum_c \frac{1}{s_c}\right)\left(\frac{2M(m)}{d^*}\right)^2, \tag{6}$$

each with its own family's m. Table 2 gives them for $k = 1, 2, 3$. By these bounds, a one-dimension difference needs 671 responses, a two-way difference 2,681, and the three-way difference 8,038. The three-way family is a single test, so its multiplier is the smallest, and the factor of four per dimension still dominates. The unequal columns split one dimension 0.25 to the rest (such as the explorer entry point) and the others 50/50, independently.

			Effect inside one cell					k-way difference of effects				
k	$\binom{D}{k}$	Cells	m	M	0.5	0.3	0.5, unequal	m	M	0.5	0.3	0.5, unequal
1	3	2	6	3.48	388	1,077	776	3	3.24	671	1,862	894
2	3	4	12	3.71	880	2,443	1,759	3	3.24	2,681	7,445	3,574
3	1	8	8	3.58	1,637	4,547	3,274	1	2.80	8,038	22,326	10,717

Table 2: Responses needed, N^*, for heterogeneity analyses of order k with $D = 3$ candidate dimensions, from equation (6). m is the number of tests in the order's family and $M = M(m)$, with the Bonferroni critical value, so the entries are upper bounds. Table 4 gives the Romano-Wolf thresholds. Columns 0.5 and 0.3 give d^* with equal shares. The unequal columns use $d^* = 0.5$ with one dimension split 0.25 to the rest.

The standard deviation σ is that of the whole control group. If a dimension explains part of the spread of the index, the spread inside a cell is smaller and these thresholds are conservative.

7 The index and its questions

The thresholds in (3) are in standardized units of each analysis's own outcome. An index and a single question have different spreads, so the same change in answers is a larger standardized effect on the index than on a question. If treatment moves every question by the same raw amount δ, each question has standard deviation σ_q, and the questions are correlated with average correlation ρ, then the index has standard deviation $\sigma_q\sqrt{\rho + (1-\rho)/K}$ and

$$d_{\text{question}} = d_{\text{index}} \times \sqrt{\rho + \frac{1-\rho}{K}}. \tag{7}$$

In the simulated population of Section 9, $\rho = 0.24$ and the factor is 0.62. An effect of 0.50 on the index is an effect of about 0.31 on each question (between 0.29 and 0.33). Detecting that on one question, with Romano-Wolf over five, takes about 390 responses in the simulations (the Bonferroni bound is 479). An effect of 0.50 on each question needs only 150 (bound 187). This is why the index is the primary outcome: it is the analysis with power at small samples.

8 What the effect sizes mean on the index

A standardized effect is the change in the mean lean index divided by the index's standard deviation in the control group. The simulated population of Section 9 has an index standard deviation of

0.50. This spread is an assumption, not a measurement. With it, an effect of d standard deviations is a change of $d \times 0.50$ in the mean index.

The index counts steps. An answer moves one step when it goes from neutral to a study's value, or from the other study's value to neutral. One step on one answer moves that respondent's index by $1/K$, one fifth. A switch from one study's value to the other's is two steps, or two fifths. The same change in the mean index can therefore come about in several ways. Table 3 gives three for each of five effect sizes, from 0.1 to 0.5. All are counted against the control group, with every move toward the same study's value.

- **0.1 standard deviations** is a change of 0.05 in the mean index. For every 100 treated respondents, that is 25 one-step moves in all, or 0.25 of a step per respondent. It is also what results if 5% of treated respondents move all five answers one step, or if 13% switch one answer from one study's value to the other's.
- **0.3 standard deviations** is a change of 0.15. That is 75 one-step moves per 100 treated respondents, or 0.75 of a step each. It is also 15% of respondents moving all five answers one step, or 38% switching one answer from one study's value to the other's.
- **0.5 standard deviations** is a change of 0.25. That is 125 one-step moves per 100 treated respondents, more than one step each (1.25). It is also 25% of respondents moving all five answers one step, or 63% switching one answer from one study's value to the other's.

Effect (SD)	Index points	One-step moves per 100 treated respondents	Share moving all five answers one step	Share switching one answer to the other value	Responses for 80% power on H1
0.1	0.05	25	5%	13%	3,140
0.2	0.10	50	10%	25%	785
0.3	0.15	75	15%	38%	349
0.4	0.20	100	20%	50%	197
0.5	0.25	125	25%	63%	126

Table 3: Five standardized effects in the lean index's own units, with the simulated index standard deviation of 0.50. A one-step move takes one answer from neutral to a study's value, or from the other study's value to neutral. The last column is N^*_{main} from (3).

9 Simulations

Data-generating process. Respondent i has a latent lean $\theta_i \sim \mathcal{N}(0, \tau^2)$. The latent answer to question k is $\theta_i + a_k + \beta T_i + \varepsilon_{ik}$ with $\varepsilon_{ik} \sim \mathcal{N}(0, 1)$, and is coded $+1$ above 0.5, -1 below -0.5, and 0 between. K fixed question intercepts a_k make some questions lean one way and some the other. $\tau = 0.66$ is calibrated so that the lean index has standard deviation 0.50. This spread is an assumption, since no answers exist yet. A single question then has standard deviation 0.81. For each panel of Figure 1, the treatment shift β is chosen so the standardized effect on the index is 0.1, 0.3 or 0.5. For 0.50 it is $\beta = 0.43$. Every sample also draws the $D = 3$ binary dimensions, independent of each other and of the strata, each split 50/50 unless stated. The estimator is the registered one: equation (1), with HC2 standard errors and t critical values for a single test. Each point is 2,000 replications (1,000 for the heterogeneity curves).

Romano-Wolf inside the simulations. Each simulated sample runs the whole procedure of Section 4: the family's regressions, 199 bootstrap draws within strata, and the step-down. The simulations use fewer draws than the registered 2,000 to save time. Fewer draws cost a little power, so the simulated thresholds lean high.

For the heterogeneity curves, the latent treatment shift differs across the halves of one dimension, or across the cells of one pair, so that the standardized k-way difference is 0.50. The other dimensions or pairs of the family carry no difference. For $k = 2$ the shift is $\pm h/4$ across the four cells of two independent 50/50 dimensions, and the test is on the coefficient of $T_i \times G_{1i} \times G_{2i}$ in the saturated version of equation (1). Each replication applies Romano-Wolf to its order's whole family: the 3 one-dimension differences, or the 3 two-way differences.

Results. Figure 1 shows the power of the main test and of a single-question test, in three panels, one for each true effect on the index. The simulated points lie on the analytic curves: at $N = 200$ and an effect of 0.50, the main test's simulated power is 94% against 94% from (2). At 0.3 standard deviations the main test's power is 54% at $N = 200$ and 100% at $N = 1{,}000$. At 0.1 standard deviations it is 11% at $N = 200$ and 39% at $N = 1{,}000$. Reaching 80% at that size takes 3,140 responses. The single question lags far behind, as (7) predicts. Its simulated points, with Romano-Wolf over the five questions, lie above the Bonferroni curve, because the questions are correlated. The probability that at least one of the five questions rejects at $N = 200$ is 80%. Under no effect, the main test rejects in between 4.5% and 5.8% of replications for every N from 30 up, so the test keeps its 5% level even with few responses.

Figure 2 does the same for heterogeneity, at a k-way difference of 0.50 standard deviations. The simulated Romano-Wolf points lie on the Bonferroni curves. The differences across independent dimensions are nearly independent tests, and only one carries an effect, so the step-down gains little here. For one dimension split 50/50, at $N = 800$, the simulated power is 86% against 87% from (6). For two dimensions, at $N = 3{,}000$, it is 87% against 85%.

9.1 Romano-Wolf thresholds

For each family and target d^*, the simulation measures power at 5 values of N, from 50% to 110% of the Bonferroni bound, with 500 replications each. Power is the share of replications in which a test with the effect is rejected, averaged over the tests with the effect. A weighted straight line through $\Phi^{-1}(\text{power})$ against $\sqrt{N}$, the shape of the normal power curve, gives the N with 80% power. That N, rounded up to the next 10, is the Romano-Wolf threshold N^* of Tables 1 and 4. Its simulation standard error is at most 3% of its value.

The effect patterns follow what each analysis looks for. For the questions and the cells, treatment shifts every respondent's latent lean by the same amount, so every test in the family has the effect: d^* on each question, or d^* on the index inside each cell. For the differences, one dimension or one pair carries a difference of d^*, and the others carry none.

For the questions and the cells, the Romano-Wolf thresholds are 70% to 83% of the Bonferroni bounds. For the differences, the tests are nearly independent and only one has an effect, so Romano-Wolf can gain little. With only 199 draws, the simulation finds no threshold below the bound in 3 of the 19 cases, and the tables keep the bound there. Where it finds one for a difference, the gain is within simulation error. The same samples, judged against the Bonferroni critical value, reach 80% power at 95% to 103% of the bound. This checks the formulas.

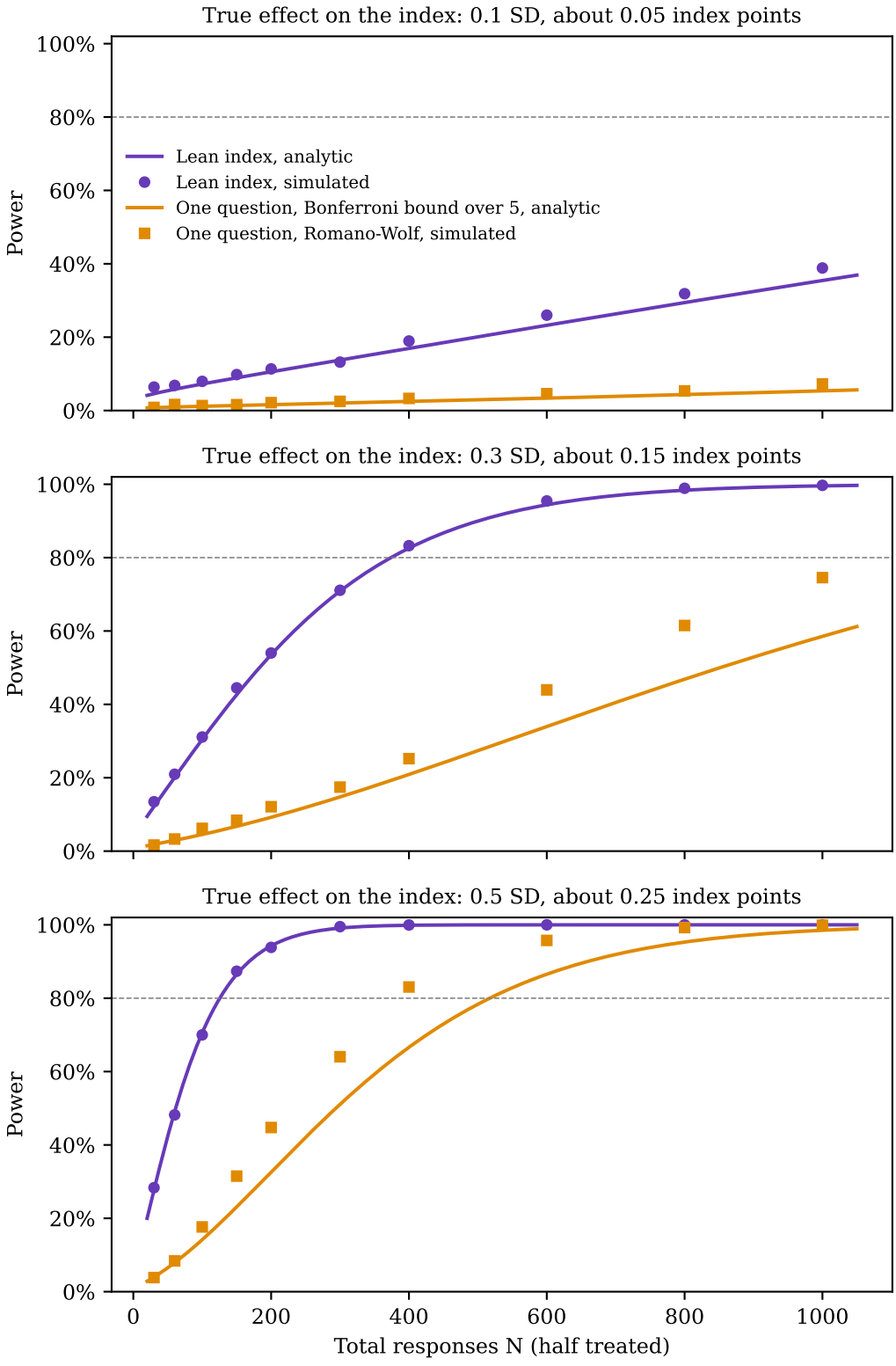


Figure 1: Power against total responses for a treatment that shifts the lean index by 0.1, 0.3 or 0.5 standard deviations (top to bottom). Table 3 gives each in index points. Lines: equation (2), with the Bonferroni bound for the single question. Points: simulations of the registered estimator, with Romano-Wolf over the five questions. The dashed line is 80% power.

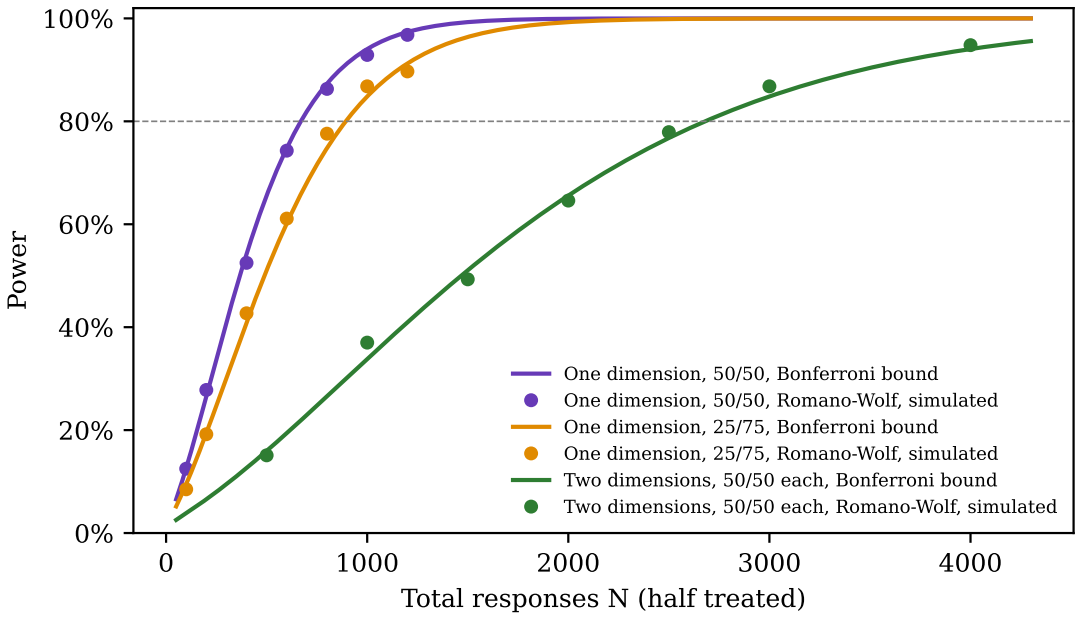


Figure 2: Power to detect a k-way difference of 0.50 standard deviations: one dimension split 50/50 or 25/75 (Romano-Wolf over 3), and two dimensions split 50/50 each (Romano-Wolf over 3). Lines: the Bonferroni bound from equation (5). Points: simulations of the registered estimator with the Romano-Wolf step-down.

10 The staged rule

Rule. At the close of collection, compute the realized number of responses N and the realized share $\hat{p}$ of each pre-treatment subgroup. Run every registered analysis whose threshold N^*, evaluated at the realized shares and at the target $d^* = 0.50$, is at most N. For a family of tests, N^* is the Romano-Wolf threshold simulated as in Section 9.1. For a single test, it comes from (3) or (6). Report every other registered analysis as "not run: below its threshold", with the threshold. The main hypothesis is always run and always reported, with its MDE at the realized N. The rule uses only N and pre-treatment shares, which carry no information about treatment effects, so applying it does not depend on the results.

Heterogeneity by order. Heterogeneity is unlocked one order at a time. Analyses of one dimension come first. Analyses of order k are run only if those of order $k-1$ of the same kind are run, and only if N reaches their own threshold. With equal shares the thresholds already rise with k, so this condition matters only when realized shares are very uneven.

The target. $d^* = 0.50$ is a medium standardized effect. A smaller target, such as 0.3, would make every threshold about $(0.5/0.3)^2 \approx 2.8$ times larger. Table 4 gives both.

What the tiers mean in practice. The numbers are the thresholds N^* of Table 4 at $d^* = 0.50$.

- Below 126 responses, only the main test is run. It can detect only effects larger than half a

Tier	Analysis	Tests m	$d^* = 0.5$		$d^* = 0.3$	
			N^*	Bound	N^*	Bound
0	H1: lean index, full sample	1	126		349	
1	Each of the five questions	5	150	187	410	520
2a	Effect inside each cell, one dimension at a time (3 combinations, 2 cells each)	6	320	388	890	1,077
2b	Difference between the two halves, one dimension at a time (H2 to H4)	3	660	671	1,862†	1,862
3a	Effect inside each cell, two dimensions at a time (3 combinations, 4 cells each)	12	690	880	1,960	2,443
3b	2-way difference of effects, two dimensions at a time	3	2,681†	2,681	7,445†	7,445
4a	Effect inside each cell, three dimensions at a time (1 combination, 8 cells each)	8	1,290	1,637	3,670	4,547
4b	3-way difference of effects, three dimensions at a time (one test)	1	8,038		22,326	
5	Each question inside each half, one dimension at a time	30	390	509	1,060	1,413

Table 4: Responses needed before each analysis is run. N^*: the simulated Romano-Wolf threshold, rounded up to the next 10. For a single test ($m = 1$), the formula. Bound: the Bonferroni bound from equations (3) and (6). †The simulation found no Romano-Wolf threshold below the bound, so the bound is kept. Heterogeneity rows assume equal shares. At the close they are recomputed at the realized shares.

standard deviation, and the write-up says so.

- From 150 responses, the five questions are each reported (tier 1). By equation (7), a question-level effect that matches an index effect of 0.50 needs about 390 responses, so question-level results between the two are reported as estimates with confidence intervals, not as tests of a common effect.

- From 320 responses, the effect inside each half of each dimension is reported (tier 2a, Romano-Wolf over 6): experts and non-experts, respondents with and without prior exposure, and the explorer entry point against the other two. If one half holds only a quarter of the responses, this needs 540. The same split by whether a respondent gives an email is descriptive only (see below).

- From 660 responses, the tests that two halves' effects differ are run (tier 2b, the plan's H2 to H4, Romano-Wolf over 3). With one half holding a quarter, 880.

- Two dimensions at a time: the effect inside each of the four cells from 690 (tier 3a), and the two-way differences from 2,681 (tier 3b).

- Three dimensions at a time: the effect inside each of the eight cells from 1,290 (tier 4a), and the three-way difference, a single test, from 8,038 (tier 4b).

- Question-by-subgroup effects, one dimension at a time (tier 5, Romano-Wolf over 30), need 1,010 responses once equation (7) is taken into account.

Email. Whether a respondent gives an email is asked at the end, after treatment, so treatment could change who gives one. Analyses split by email are therefore descriptive. They are reported only together with the treatment's effect on the share who give an email.

Other dimensions. A dimension added later raises D, which enlarges every family and every threshold above. It must be registered before the data are seen, and equation (6) and the simulations are then recomputed with the new D. A split on a question asked after treatment is descriptive only.

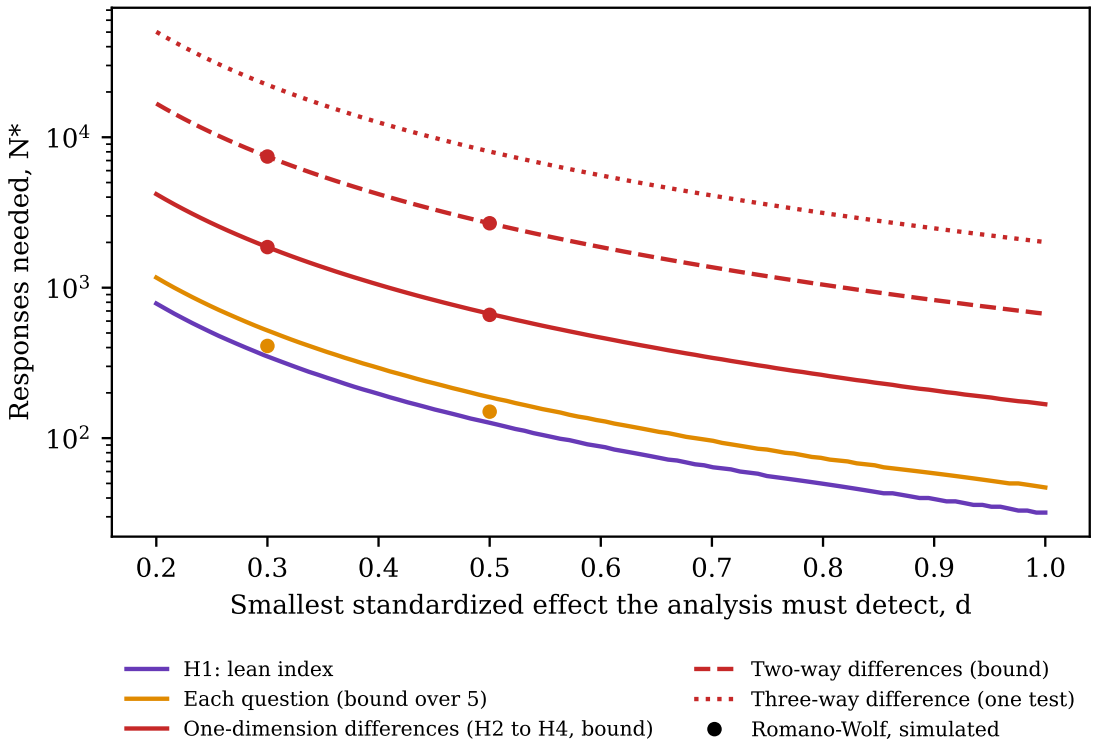


Figure 3: Responses needed, N^*, against the effect the analysis must detect, with equal shares. Lines: equations (3) and (6), Bonferroni bounds for the families. Points: the simulated Romano-Wolf thresholds at $d = 0.3$ and 0.50. Log scale.

11 Limits

The thresholds assume equal arms, independent responses, and standardized effects. Unequal arms raise them slightly. The simulated population is one plausible shape of answers, chosen to match the assumed spread of the index. If answers are more correlated across questions, the gap between the index and single questions narrows. If they are less correlated, it widens. None of this affects the validity of the main test, whose level the simulations confirm at every sample size. It affects only how large the effects must be to be seen. The Romano-Wolf thresholds also depend on the assumed effect patterns and correlations. The Bonferroni bounds do not.

References

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Romano, J. P. and M. Wolf (2005). Stepwise multiple testing as formalized data snooping. *Econometrica* 73(4), 1237–1282.